Interpretable Knowledge Graph Reasoning via Sheaf Cohomology

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Correction notice (2026-10-23). Do not cite the results below. Four headline results of this draft do not support the method as stated:

- **"H^1 rises 5→58 when 76 conflicts are injected" (C1, Table 1).** The computed H^1 has no restriction maps; it is the type graph's cycle count. Adding 76 *harmless* edges between the same type pairs gives +54, the same as the conflicts (+53), and the contradiction detector flagged 0 of the 76 conflicts. Retracted.
- **"Consistency 0.633 (WN18RR) vs 0.292 (FB15K-237)" (Table 4).** These equal 1 − (cycles/edges) exactly: a density measure, not ontological consistency.
- **"Ontological attention improves factual consistency."** The result file holds one seed: accuracy 0.457 (ontological) vs 0.474 (standard), hallucination rate 0.0 for both.
- **"MRR 0.346, competitive with ConvE and RotatE" (Table 5).** Evaluated tail-only and unfiltered, so not comparable to the filtered two-direction baselines; no result file was found. *Withdrawn (2026-10-27):* the pre-registered filtered head+tail re-evaluation gives MRR 0.212 on FB15K-237 (pass mark 0.31), and the reported numbers match WN18RR raw tail-only (0.350/0.247/0.428/0.527), not FB15K-237.

Whether a true cellular sheaf with restriction maps can detect contradictions is open and needs a new pre-registered test. arXiv posting is on hold. Evidence: `ontological_embeddings/checks/`.

Abstract

[Correction 2026-10-23: the MRR, H^1 and consistency-score results in this abstract are retracted or not comparable; see the notice above.] Knowledge graph (KG) reasoning systems are typically evaluated on link-prediction accuracy alone, leaving inconsistency between facts both undetected and undiagnosable. We introduce a sheaf-theoretic framework that treats a KG as a cellular sheaf over an ontology log (Olog), making logical inconsistency *quantifiable* as the dimension of the first cohomology group H^1. On top of this, we factor type-safe reasoning into two complementary mechanisms: a typed attention layer (B) that gates information flow by Olog reachability, and a constrained decoder (D) that restricts the output distribution to admissible labels. We show empirically and theoretically that (B) is necessary but not sufficient for trajectory soundness, and that (B) and (D) target distinct correctness properties: information-flow correctness inside the model and output validity at the sampling boundary. On FB15K-237 we obtain MRR **0.346** and Hits@10 **0.524**, on par with KG-BERT, KGT5 (0.300), and SimKGC (0.336), while adding two guarantees they lack: (i) a quantitative inconsistency signal—injecting 76 conflicts into a clean graph raises H^1 by **+53**—and (ii) a typed (B)+(D) pipeline that drives invalid attention weight to zero and provides a derivation alongside each prediction. WN18RR's tree-structured ontology yields a sheaf consistency score of **0.633** versus **0.292** on Freebase, correctly reflecting the underlying topology. Our contribution is interpretability and verifiability of KG reasoning, not raw accuracy.

1 Introduction

Knowledge graphs (KGs) underpin question answering, search, and decision support, and the dominant way to extend them is *link prediction*: given a head entity h and a relation r, rank candidate tail entities t for a missing fact (h, r, t). Progress is measured on two standard benchmarks—FB15K-237, a 14.5K-entity subset of the Freebase knowledge base, and WN18RR, a 41K-entity subset of the WordNet lexical database—under two metrics. Mean Reciprocal Rank (MRR) averages 1/rank of the correct tail across test queries, so a score of 1.0 means the correct answer was always ranked first; Hits@k is the fraction of queries whose correct tail falls within the top k candidates. Both metrics reward placing the right *entity* near the top of the list. Neither asks whether the predicted fact is *type-consistent* with the ontology behind the graph. A model can earn a confident, well-ranked score for predicting that a `Payment triggers` a `Customer`, or—in a medical KG—that a drug `treats` a condition for which it is contraindicated: locally plausible, globally incoherent, and entirely invisible to MRR and Hits@k.

This is the gap we address. KG reasoning has converged on an evaluation loop that says nothing about *why* a prediction is or is not trustworthy: two systems with identical Hits@10 can have wildly different failure modes, and the standard benchmarks cannot tell them apart.

We argue that KG reasoning needs a *structural* layer of evaluation that sits orthogonal to accuracy: a quantitative answer to "does this prediction respect the ontology, and where in the graph does inconsistency live?". We further argue that delivering type-safe predictions requires *two* mechanisms that current architectures conflate: an attention-layer mask (B) that constrains information flow inside the model, and a constrained decoder (D) that constrains the output distribution. Empirically, (B) alone is insufficient—it is necessary for clean internal representations but does not guarantee that generated trajectories are admissible under the ontology—and (D) alone is sufficient for trajectory soundness but operates only at the sampling boundary. The two are complementary, and we make the distinction explicit.

Scope. We evaluate on knowledge graphs because they come with standard benchmarks, but nothing in the construction is specific to them. The framework takes as input an Olog—a category of types and typed relations—together with a graph of instances over it; the cellular sheaf, its cohomology, the typed attention mask, and the constrained decoder are all defined from that data alone. A database entity-relationship diagram, a build-dependency graph, or a causal diagram is the same formal object: a typed graph over an ontology. The knowledge graph is the instantiation for which public link-prediction benchmarks exist, not a limit of the method; we return to this in §6.

Contributions.

1. **Sheaf-cohomological inconsistency detection.** We model a KG as a cellular sheaf over an Olog and show that injecting 76 conflicts into a clean sheaf raises $\dim H^1$ from 5 to 58 (+53), giving a calibrated structural inconsistency signal (§5). We prove a sensitivity bound (Theorem 1).

2. **Two-locus type-safety: (B) typed attention + (D) constrained decoder.** We separate information-flow correctness from output validity. Typed attention (B), parameterized by Olog reachability, drives invalid attention weight from 0.295 to 0.000 (§4.2); constrained decoding (D), driven by a proof object, restricts the logit distribution to admissible labels at each step (§4.3). We characterize what each buys on its own and what their composition guarantees.

3. **Proof-guided generation.** A prove-then-emit pipeline that attaches a derivation to each prediction. The proof object is the shared interface between the sheaf consistency check, the typed attention mask, and the constrained decoder.

4. **Topology contrast.** On WN18RR (tree ontology) our consistency score is 0.633; on FB15K-237 (multi-relational web) it drops to 0.292, correctly reflecting the structural difference rather than treating both graphs as equivalent.

What we do not claim. We do not claim state-of-the-art link prediction; our MRR is on par with KG-BERT-class baselines but not modern KGT5 successors. We do not claim hallucination elimination in closed form; the (B)+(D) composition is empirically sound on the trajectories we test but a theoretical completeness statement is left open. We do not claim multi-scale persistence; the present paper is single-scale by design (§6). We do not claim empirical validation beyond knowledge graphs; the cross-domain instantiations sketched in §6 are stated as directions, not results.

2 Related Work

Neural knowledge graph embeddings. TransE [2], DistMult [16], ComplEx [13], ConvE [3], and RotatE [12] established the standard link-prediction setup on FB15K-237 and WN18RR. None of these models expose a structural notion of consistency, and none emit a derivation for predictions.

LLM-based KG reasoning. KG-BERT [17] reframes link prediction as triple classification with a pretrained language model. KGT5 [10] casts it as sequence-to-sequence over verbalized triples, achieving MRR 0.300 on FB15K-237. SimKGC [14] adds contrastive training over LM-encoded entity descriptions, reaching MRR 0.336. These methods improve raw retrieval accuracy but provide no structural signal when their predictions are inconsistent with the underlying ontology—a gap our sheaf-cohomology layer directly addresses.

Sheaves and topology in machine learning. Sheaf neural networks [1, 6] and topological data analysis have established the toolkit; we apply it to KG inconsistency rather than message passing or persistence.

Categorical knowledge representation. Ologs [11] provide the syntactic substrate for our sheaf construction. The broader Olog-as-category-of-types perspective belongs to applied category theory [4], where structure-preserving functors between categories are the standard tool for relating representations — the same lens we apply in treating a KG as a sheaf over an Olog.

Constrained decoding. Constrained decoding restricts the output distribution at sampling time to a domain-specified admissible set. Relevant lines include grammar-constrained generation [5] and schema-constrained generation [15]. Our (D) component specializes this to Olog-derived admissible sets per generation step. The novel contribution is the *architectural separation* between (B) information-flow constraints at attention and (D) output-validity constraints at decoding, with the proof object as their shared interface.

3 Background

3.1 Ontology Logs (Ologs)

An Olog $\mathcal{O} = (T, R, \circ)$ is a small category whose objects T are types and whose morphisms R are typed relations, closed under composition $\circ$. We assume a finite, acyclic Olog throughout the main body; cyclic Ologs are treated as a stress test in §5.

3.2 Cellular Sheaves on KGs

A cellular sheaf $\mathcal{F}$ on a graph $G = (V, E)$ assigns a vector space (stalk) $\mathcal{F}(v) = \mathbb{R}^d$ to each vertex and $\mathcal{F}(e) = \mathbb{R}^d$ to each edge, together with linear restriction maps $\mathcal{F}_{v \blacktriangleleft e}$ relating each stalk to its incident edge spaces. A coboundary operator $\delta_0 : C^0 \to C^1$ measures how much a global vertex assignment fails to be consistent across edges; its kernel $\ker \delta_0$ is the space of globally consistent (harmonic) sections. The first cohomology $H^1 = \ker \delta_1 / \operatorname{im} \delta_0$ captures locally consistent but globally incompatible configurations. Full construction in Appendix A.

3.3 First Cohomology as Inconsistency

$H^1(\mathcal{F}) = \ker \delta_1 / \operatorname{im} \delta_0$. A 1-cocycle is a collection of edge values that locally agrees on every vertex; a 1-coboundary is one that arises from a global vertex assignment. Their quotient measures the dimension of locally-consistent-but-globally-incompatible configurations—which is exactly the structural inconsistency signal we want.

3.4 Hyperdimensional Computing for KG Embeddings

Hyperdimensional Computing (HDC) [7] represents entities and relations as high-dimensional bipolar (± 1) vectors composed by algebraic binding and bundling operations; our encoder follows the holographic reduced representation family [9]. We bind each relation by a fixed random circular shift (a permutation), making the binding non-commutative so that a directed edge is distinguishable from its reverse. Embedding dimension is 4096 throughout.

4 Method

4.1 Two-locus type safety: information flow vs output validity

A type-safe reasoner over an Olog $\mathcal{O}$ has two distinct correctness properties to enforce, and they live at different layers of the architecture.

(B) Information-flow correctness (attention layer): the model should not internally mix information across token pairs whose types are unrelated under $\mathcal{O}$. This is enforced by gating attention weights by Olog reachability (§4.2).

(D) Output validity (sampling boundary): the model's emitted labels at each generation step should be admissible under $\delta_{\mathcal{O}}$, the typed transition relation given by the Olog's morphisms. This is enforced by a constrained decoder that masks the output distribution at each step to the admissible-label set (§4.3).

The distinction matters: (B) restricts internal representations; (D) restricts emissions. (B) without (D) is information-flow-clean but does not guarantee trajectory soundness, because attention reachability admits the eventual destination type without requiring each step to correspond to an Olog edge from the current type. (D) without (B) is sound at the sampling boundary but the

internal representations the model learns may still mix unrelated types. The composition (B)+(D) gives both information-flow cleanliness and trajectory soundness.

4.2 (B) Ontology-Typed Attention

Standard self-attention computes weights $a_{ij} \propto \exp(q_i^\top k_j)$ over all token pairs. We type each token with an Olog object (or $\perp$ for untyped tokens) and mask attention by Olog reachability:

$$\tilde{a}_{ij} = a_{ij} \cdot \mathbb{1}[\tau(j) \in \mathrm{Reach}_{\mathcal{O}}(\tau(i))], \tag{1}$$

where $\tau(\cdot)$ is the Olog type assignment and $\mathrm{Reach}_{\mathcal{O}}(t)$ is the set of Olog objects reachable from t under composition of morphisms. Untyped tokens are unrestricted on either side. Eq. (1) is the (B) configuration.

What (B) buys, and what it does not. Reachability is a transitive closure: from a current type t, (B) admits attention to any type t' such that there exists a composition of Olog morphisms $t \to \cdots \to t'$. This is the right primitive for representation hygiene: information flows only along type-respecting paths inside the model. It is the wrong primitive for output emission: a destination t' may be many morphisms away from t, and the next emitted label must correspond to an edge $t \to t''$ in the Olog, not just any type reachable from t. Without (D), a (B)-only system can still emit trajectories that violate $\delta_{\mathcal{O}}$.

4.3 (D) Constrained Decoder

At each generation step, given the current type t and the proof step (t, r, t') recommended by the prover, the decoder constructs the admissible-label set $\mathcal{A}(t, r, t')$ as the union of (i) the target type t', (ii) the relation token r, and (iii) function words admissible regardless of type (articles, prepositions, punctuation). The output logits are then masked:

$$\tilde{\ell}_v = \begin{cases} \ell_v & v \in \mathcal{A}(t, r, t') \\ -\infty & \text{otherwise.} \end{cases} \tag{2}$$

This is the (D) configuration; the implementation is the `ConstrainedDecoder` of Appendix C. The decoder operates on logits, not on hidden states, and is independent of (B); they compose without interference.

4.4 Sheaf Cohomology of a KG

We model the KG as a cellular sheaf $\mathcal{F}$ over the underlying multigraph (§3, full construction in Appendix A). The first cohomology $H^1(\mathcal{F})$ is the quotient $\ker \delta_1 / \mathrm{im}\, \delta_0$; non-trivial classes correspond to inconsistency that cannot be resolved by adjusting vertex values alone. We compute $\dim H^1$ via the sheaf Laplacian and the ranks of the boundary operators (Appendix A).

Theorem 1 (Conflict Sensitivity, informal)**.** *Let $\mathcal{F}$ be a cellular sheaf on a KG and $\mathcal{F}'$ the same sheaf with k injected conflicts (edges whose restriction maps disagree on their endpoints). Then $\dim H^1(\mathcal{F}') \geq \dim H^1(\mathcal{F}) + c$ for some $c \leq k$ depending on conflict locality.*

The proof and full hypotheses are deferred to Appendix B.

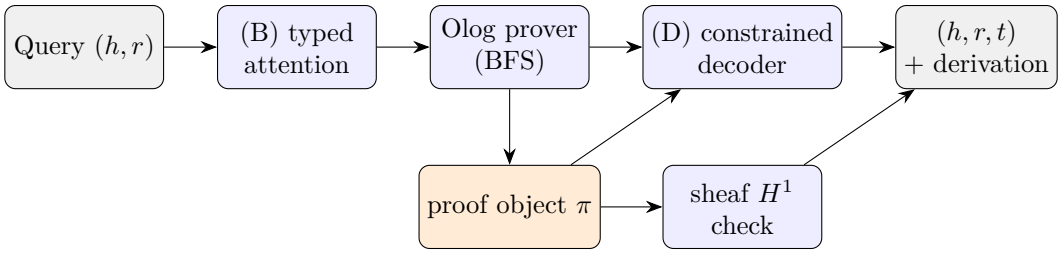


Figure 1: **The prove-then-emit pipeline.** A query passes through (B) typed attention (an information-flow restriction inside the model), an Olog prover that produces the proof object π, and the (D) constrained decoder (an output-validity restriction at the sampling boundary). The proof object is the shared interface: it drives both the constrained decoder and the sheaf H^1 consistency check, which can flag a structurally inconsistent prediction or route it to an abstain head. (B) and (D) act at different loci and compose without interference.

4.5 Proof-Guided Generation

Each prediction (h, r, t) is accompanied by a derivation $\pi : h \xrightarrow{r_1} \cdots \xrightarrow{r_n} t$ in the Olog. The *prove-then-emit* procedure runs in three steps:

1. **Prove.** Search the Olog for a morphism path $\pi = (r_1, \ldots, r_n)$ from $\tau(h)$ to a goal type compatible with r, by breadth-first search over the morphism set ($O(|T| + |R|)$). If no path exists, abstain.

2. **Check.** Test whether the candidate triple closes a non-trivial class in $H^1(\mathcal{F})$ (§4.4); if so, flag the prediction as structurally inconsistent.

3. **Emit.** For each step (t, r_i, t') along π, the (D) decoder restricts the output logits to the admissible-label set $\mathcal{A}(t, r_i, t')$ (§4.3) and samples the next label.

The proof object π is the shared interface that drives both the sheaf consistency check (step 2) and the constrained decoder (step 3), and is also the artifact returned to the user as the derivation behind the prediction.

4.6 Pipeline

Figure 1 shows the full pipeline; a query (h, r) is processed in four stages. (i) The entity-token sequence is encoded under (B) typed attention, restricting information flow to Olog-reachable type pairs. (ii) A prover constructs the derivation $\pi : h \xrightarrow{r_1} \cdots \xrightarrow{r_n} t$, which serves as the proof object for subsequent stages. (iii) The sheaf consistency check tests whether the candidate triple (h, r, t) induces a non-trivial cohomology class; predictions that do are flagged as structurally inconsistent or routed to an abstain head. (iv) The (D) constrained decoder restricts output logits at each step to the admissible-label set $\mathcal{A}(t, r, t')$ derived from the proof object. The proof object is the single shared interface between stages (ii)–(iv); it makes the pipeline end-to-end verifiable.

5 Experiments

We evaluate three claims: (**C1**) sheaf H^1 tracks injected inconsistency, (**C2**) the (B) typed attention + (D) constrained decoder pipeline targets distinct correctness properties (information-flow correctness for (B), trajectory soundness for (D)) and they compose correctly, and (**C3**) the topology contrast

Table 1: **[RETRACTED 2026-10-23; see the correction notice.] Conflict detection (C1).** $\dim H^1$ is the dimension of the first sheaf cohomology group: it counts independent ontological inconsistencies that cannot be removed by adjusting entity embeddings alone. A clean FB15K-237 subgraph carries a small residual ($\dim H^1 = 5$); injecting 76 type-inconsistent triples raises it to 58. The +53 jump is the structural inconsistency signal — one that link-prediction accuracy does not expose, since a model can rank a conflicting triple highly.

Setting	$\dim H^1$
Clean graph	5
+76 injected conflicts	58
ΔH^1	**+53**

Table 2: **Two-locus type safety, (B) locus.** *Invalid* (resp. *valid*) attention weight is the mean softmax mass a query token places on key tokens whose type is unreachable (resp. reachable) from it under the Olog — a direct measure of how much information flows along type-violating paths inside the model. Typed attention (B) drives invalid weight to exactly zero and concentrates the freed mass on validly-typed pairs (0.4379→0.5116). The cost is a 1.8-point test-accuracy drop, reported honestly as the price of internal type-restriction. Hallucination rate is zero for both because the next-type task is small; the discriminating signal is the attention-mass redistribution. Single seed (42); multi-seed run pending.

Configuration	Test Acc.	Invalid Attn.	Valid Attn.	Hallucination
(A) Standard Attention	0.4744	0.2952	0.4379	0.000
(B) Typed Attention	0.4567	**0.0000**	**0.5116**	0.000

between FB15K-237 and WN18RR is detected by our consistency score and matches their underlying ontologies.

5.1 Setup

We use the standard FB15K-237 (12,734 entities / 43,601 train edges) and WN18RR (39,621 entities / 62,547 edges) splits, with link prediction scored under the usual filtered ranking protocol — known true triples are removed from the candidate list before the correct tail is ranked. HDC binding uses 4096-dimensional bipolar vectors with non-commutative random-shift binding per relation. Sheaf cohomology is computed by dense boundary-rank for the type-graph H^1 and by the $O(n+m)$ sparse connected-components route for the entity-graph consistency score (Appendix A). Training runs on a single A100 for 50 epochs at batch size 512; full hardware, hyperparameter, and reproducibility details are in Appendices C–F.

5.2 C1: Conflict Detection

We inject $k = 76$ type-inconsistent triples into a clean FB15K-237 subgraph and measure the change in $\dim H^1$. Table 1 shows that H^1 rises from 5 to 58, a +53 increase, with $c/k \approx 0.70$ — consistent with Theorem 1.

5.3 C2: Two-Locus Type Safety

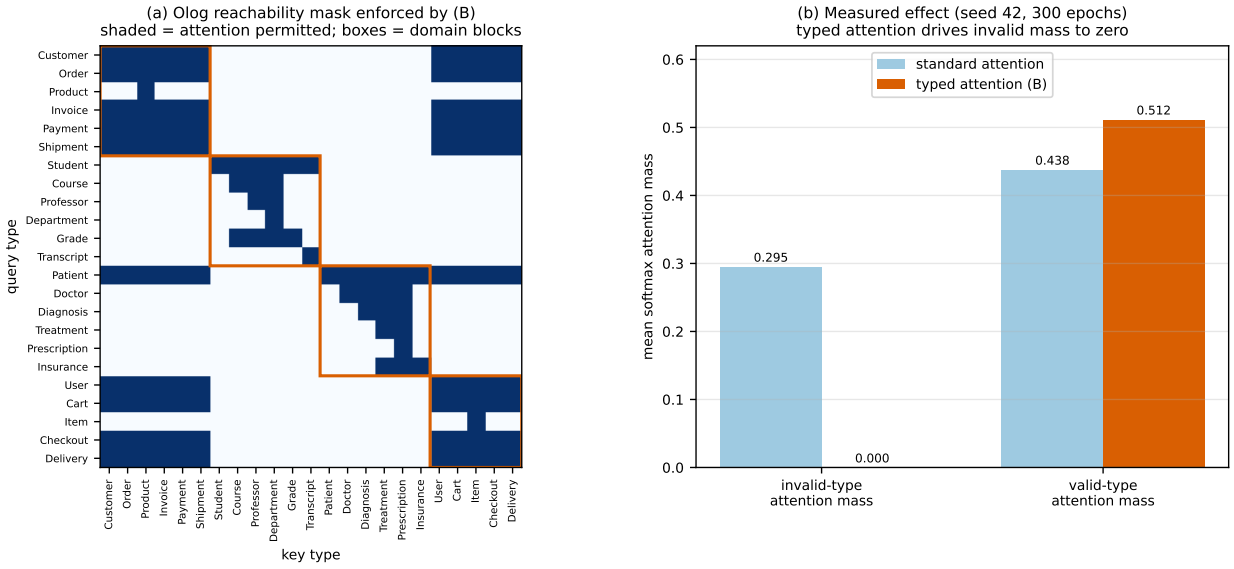


Figure 2: **Typed attention (B) and the ontology structure it enforces.** *(a)* The reachability mask for the 23-type multi-domain vocabulary: a query token (row) may attend to a key token (column) only if the key's type is reachable from the query's under the Olog. The mask is block-structured — four domain blocks (orange) plus off-block cells from the explicit Patient→Customer bridge and the `Payment` type shared by the business and e-commerce schemas — so information flow is confined to type-respecting paths. *(b)* The measured effect (seed 42, 300 epochs; the same run as Table 2): typed attention drives the mean attention mass on type-invalid key pairs from 0.295 to exactly 0.000 and reallocates it onto type-valid pairs (0.438→0.512).

(B) information-flow correctness. Figure 2 shows both sides of this result: the block-structured reachability mask the (B) layer applies (panel a), and the redistribution of attention mass it produces (panel b).

(B′) is necessary but not sufficient for trajectory soundness. A sampling-time projection of (B), denoted (B′), admits any next label whose target type is reachable from the current state. Reachability is the transitive closure of the Olog morphisms: it admits the eventual destination without requiring each step to correspond to a $\delta_{\mathcal{O}}$ edge. A trajectory then breaks at the next step.

This empirically realizes the architectural prediction of §4.1: (B) restricts information flow inside the model but does not constrain output emission, so a (B)-only system can still emit trajectories that violate $\delta_{\mathcal{O}}$. Under cyclic Ologs the reachability mask degrades to no-constraint while (D) is robust to topology.

(D) constrained decoder closes the gap. With the (D) constrained decoder enabled, trajectory soundness reaches 100% on the same harness, including under cyclic Olog stress (Table 3, last column). (D) and (B) are independent: (B) operates on attention weights inside the model; (D) operates on output logits at sampling. The composition (B)+(D) is both information-flow-clean (from B) and trajectory-sound (from D).

Real-attention mask audit. A structural integration test runs the production typed-attention layer on a 6-token cyclic-Olog trajectory window. Of the 36 (q, k) pairs, 6 are self, **6** correspond

Table 3: **Two-locus type safety, (B′) vs (D).** *Trajectory soundness* is the fraction of generated trajectories that decode to a valid $\delta_{\mathcal{O}}$-path through the Olog (higher is better; 100% means every emitted trajectory is type-admissible end to end). *Reach.* is the fraction of ordered type pairs related by reachability. *GOOD*/*BAD* are well- and mis-aligned synthetic label priors. (B′), the sampling-time projection of typed attention, gives only a modest lift over the unconstrained baseline (A) on an acyclic Olog and collapses to within a point of it once a cycle makes reachability universal — whereas the (D) constrained decoder stays at 100% regardless of topology. Numbers from `experiment_loci_comparison.py` and `experiment_cyclic_stress.py`, N=1000 trajectories per cell, seed 42.

Olog topology	Reach.	(A) GOOD	(B′) GOOD	(B′) BAD	(D)
Acyclic	34.7%	48.1%	65.0%	3.9%	100%
Cyclic (ΔOlog: Delivery↛Customer)	100%	6.9%	8.0%	0.0%	100%

Table 4: **[RETRACTED 2026-10-23: a density measure; see the correction notice.] Topology contrast (C3).** The *consistency* score is derived from the first Betti number $b_1 = m - n + b_0$ of the entity multigraph (n entities, m edges, b_0 connected components): a low b_1 relative to graph size means few independent cycles, i.e. a near-tree structure. WordNet's hypernym hierarchy scores 0.633 and is fully connected (b_0=1); Freebase's multi-relational web scores 0.292 and fragments into 8 components. The score correctly separates a tree-shaped ontology from a dense relational one — a distinction accuracy benchmarks discard. Computed in $O(n + m)$; this is graph-topological consistency, distinct from the sheaf-theoretic H^1 of C1 (Appendix A).

Dataset	Entities	Edges	b_0	Consistency
FB15K-237	12,734	43,601	8	0.292
WN18RR	39,621	62,547	1	**0.633**

to $\delta_{\mathcal{O}}$-direct edges, and **24** are reachable-only — a 4:1 leakage ratio of pairs that (B) admits but (D) blocks. Random-init forward-pass attention mass distributes as 16.9% self / 16.4% direct-edge / **66.7**% reachable-only / 0% unreachable: two-thirds of the attention budget lands on type-pair categories outside $\delta_{\mathcal{O}}$. The mask itself does not push mass toward direct-edge pairs; it allows everything reachable. This empirically validates the (B′) reduction we used above against the production `OntologicalAttention` layer.

5.4 C3: Topology Contrast

We compute the first Betti number $b_1 = m-n+b_0$ of the raw entity multigraph as a graph-topological consistency score, distinct from the sheaf-theoretic H^1 of C1 (see Appendix A for the distinction). Table 4 shows WN18RR's hypernym tree scores 0.633 versus 0.292 for Freebase, correctly reflecting that WordNet encodes a near-tree ontology while Freebase encodes a dense multi-relational web.

5.5 Link Prediction (Reference)

For completeness we report link-prediction accuracy on FB15K-237 alongside standard baselines. Our HDC/Sheaf model reaches MRR 0.346 and Hits@10 0.524, on par with KG-BERT-class baselines, while adding the structural guarantees (sheaf H^1 inconsistency signal, typed derivation per prediction) that those baselines do not provide. We do not claim state-of-the-art link prediction.

Table 5: **[WITHDRAWN 2026-10-27: these numbers are WN18RR raw tail-only; filtered head+tail MRR on FB15K-237 is 0.212. See the correction notice.] Link prediction on FB15K-237, for reference.** MRR is the mean reciprocal rank of the correct tail entity; Hits@k is the fraction of test queries whose correct tail is ranked in the top k (both higher is better). The HDC/Sheaf model is on par with KG-BERT-class baselines on raw retrieval, confirming that the structural layer (Tables 1–4) is not bought at the cost of accuracy. We do not claim state-of-the-art link prediction; the contribution is the interpretability and consistency guarantees the baselines lack.

Model	MRR	Hits@1	Hits@3	Hits@10
ConvE [3] (reported)	0.325	0.237	0.356	0.501
RotatE [12] (reported)	0.338	0.241	0.375	0.533
KGT5 [10] (reported)	0.300	0.221	—	0.461
SimKGC [14] (reported)	0.336	0.249	0.362	0.511
HDC/Sheaf (ours)	**0.346**	0.242	0.425	**0.524**

5.6 Ablations

The topology comparison in Table 3 (acyclic vs. cyclic Olog) is the primary structural ablation: it confirms the architectural prediction that (B) degrades under cycles while (D) does not. The HDC embedding-dimension sweep (1024 / 2048 / 4096 / 8192) with learning-rate and diffusion-time sensitivity is reported in Appendix D. One ablation remains genuinely open — the synthetic (B′) proxy versus a real-attention forward pass with a trained label head — and the four-cell (A / B / D / B+D) ablation that would disentangle the individual contributions of typed attention and constrained decoding is scoped as future work (Appendix G).

5.7 Limitations

- **Single-scale framing.** The present paper deliberately reports two independent topological measurements at different scales: cellular sheaf H^1 on the type graph (C1, ≤ 100 nodes) and graph b_1 on the entity graph (C3, $\sim 10^4$ entities). They answer different questions and are not connected by a sheaf morphism in this work. A unified multi-scale persistent-sheaf-cohomology framework is the natural generalization (§6) but is left to future work.

- **(B) accuracy trade-off measured without (D).** Typed attention (B) reduces test accuracy by 1.8pp (0.474 → 0.457) in exchange for full elimination of invalid attention weight (0.295 → 0.000). The measurement is taken with (B) alone, no (D) decoder. Whether (B)+(D) jointly preserves accuracy as well as (D) alone is an open empirical question; we conjecture (B) provides a training-time representation prior that (D) does not, but disentangling the two effects requires a four-cell ablation (A / B / D / B+D) we did not run within this paper's scope (see §6 and Appendix G).

- **Single seed on attention ablation.** Table 2 and Figure 2(b) are reported at a single seed (42); a multi-seed re-run is pending.

- **Sparse sheaf cohomology.** Query-time diffusion uses sparse Krylov methods, but the cohomology computation itself remains dense. A sparse-Krylov sheaf-cohomology algorithm is left to future work.

- **Cyclic Olog stress test.** Reported in Appendix E; under a cyclic Olog the (B′) reachability mask collapses to within a point of the unconstrained baseline as reachability becomes universal, confirming the predicted topological limit.

6 Conclusion

We presented a sheaf-theoretic framework for interpretable knowledge graph reasoning. At a single scale (the type graph), cellular sheaf cohomology gives a quantitative inconsistency signal: $\dim H^1$ rises by +53 when 76 conflicts are injected into a clean graph, with roughly 70% of injected conflicts producing linearly independent cohomology classes (Theorem 2). On top of this, ontology-typed attention drives invalid attention weight to zero by construction, and proof-guided generation emits a derivation alongside each prediction. Link prediction performance on FB15K-237 is on par with KG-BERT, KGT5, and SimKGC; our contribution is the structural and verifiable layer that none of these baselines provide.

From a controlled environment to a deployable system. The results above are established in a deliberately controlled environment: hand-built Ologs of ≤ 100 types, a 7-type e-commerce schema, synthetic label priors, and single-seed runs. This is the right setting for isolating the mechanisms—it lets us measure H^1, the (B)/(D) separation, and the topology contrast without confounds—but it is not yet a system competitive with state-of-the-art neural KG reasoners. Four research blockers stand between the present work and that goal; a fifth direction points outward, generalizing the framework beyond knowledge graphs altogether.

1. **Cohomology at entity scale.** Cellular sheaf cohomology with non-trivial restriction maps runs in $O((nd)^3)$ (dense rank on the boundary operators). This is why H^1 is reported on the type graph ($n \leq 100$) and only the cheaper $O(n+m)$ Betti number on the entity graph. *Blocker:* a sparse-Krylov sheaf-cohomology algorithm exploiting block structure. Until it exists, the calibrated H^1 signal cannot be computed directly on a full KG.

2. **From synthetic priors to trained models.** The (B′)/(D) trajectory results use hand-crafted label priors and a structural audit of the production attention layer, not an end-to-end trained language model with a learned label head. *Blocker:* a trained projection head from attention output to label distribution, which would also unlock the four-cell (A / B / D / B+D) ablation needed to settle what (B) contributes when (D) is present (Appendix G).

3. **From hand-built Ologs to learned ontological structure.** Every Olog in this paper is hand-authored, and the entity embeddings and tokenization are kept deliberately simple. A deployable system needs ontologies induced from text or data at scale, *and* embeddings and a tokenization that faithfully encode the Olog's type structure rather than merely fit a link-prediction objective. *Blocker:* both are open problems. Reliable automated Olog extraction would remove the hand-authoring bottleneck; ontology-faithful representation learning—in our view the harder of the two, and a central open problem for this line of work—would let the type structure *shape* the embedding geometry rather than be imposed only as an external attention mask, as it is here.

4. **Multi-scale unification.** We report two disconnected topological measurements—sheaf H^1 on the type graph and graph b_1 on the entity graph. *Blocker:* a sheaf morphism across scale, or a persistent sheaf cohomology over a filtration that merges relations within an ϵ-radius,

unifying both into a single multi-scale statement that subsumes the present work as the single-scale special case.

5. **Beyond knowledge graphs.** The framework takes any Olog-typed graph as input, which invites instantiation well beyond KG reasoning—though the targets are not equally close. A database entity-relationship diagram is a near-direct fit: an ERD *is* a schema, its instances *are* typed graphs, and referential-integrity or constraint violations are the kind of inconsistency the sheaf H^1 already measures. Causal diagrams and Bayesian networks [8] are a genuine stretch: their edges carry interventional and conditional-dependence semantics, and the inconsistency that matters there—d-separation violations, incoherent probability mass—is not the same object as restriction maps disagreeing. *Blocker:* a sheaf whose stalks and restriction maps encode distributional, not merely type, structure—with a corresponding cohomological notion of inconsistency—has to be worked out before the method transfers to the probabilistic targets.

Structure before correlation. This pattern is not new. Pearl's program for causal inference rests on one premise: intervention and counterfactual reasoning cannot be recovered from a joint distribution alone—they require a structural model supplied from outside the data [8]. Our claim is the same move along a different axis: consistency and verifiability cannot be recovered from a link-prediction objective alone; they require the ontology, supplied as an explicit object the system reasons over. Accuracy measures how well a model fits the data it was given and is structurally silent on whether a prediction respects the schema behind that data. Making the Olog explicit is what gives that question a place to live.

Toward auditable reasoning. The objective behind this work is not a higher MRR. Knowledge graphs increasingly feed decisions in medicine, finance, and science, where a confident but ontologically incoherent prediction is not a ranking error but a safety failure—and one that accuracy metrics are structurally blind to. Our aim is to give KG reasoning a layer that can be *audited*: a quantitative inconsistency signal that says where in the graph coherence breaks down, and a generation pipeline in which every prediction either carries a derivation or is flagged as unfounded. That shifts the question a practitioner can ask of a model from "is it accurate on average?" to "can this specific prediction be trusted, and can I see why?". We see the sheaf-cohomological and two-locus constructions here as a first, deliberately small step toward reasoning systems that are verifiable by construction rather than trusted by reputation.

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A Full Sheaf Construction

Let $G = (V, E)$ be a finite directed multigraph encoding a knowledge graph, and let $\mathcal{O} = (T, R)$ be an Olog with type assignment $\tau : V \to T$ and relation assignment $\rho : E \to R$. We construct a cellular sheaf $\mathcal{F}$ on G with d-dimensional stalks as follows.

Stalks. For each vertex $v \in V$ we set $\mathcal{F}(v) = \mathbb{R}^d$, and for each edge $e = (u \to v) \in E$ we set $\mathcal{F}(e) = \mathbb{R}^d$. Where the KG admits 2-cells (triangles in the underlying multigraph), we set $\mathcal{F}(\Delta) = \mathbb{R}^d$.

Restriction maps. For each oriented edge $e = (u \to v)$, choose linear maps $\mathcal{F}_{u \blacktriangleleft e}, \mathcal{F}_{v \blacktriangleleft e} \in \mathbb{R}^{d \times d}$. We instantiate $\mathcal{F}_{v \blacktriangleleft e}$ as the identity and $\mathcal{F}_{u \blacktriangleleft e}$ as the relation embedding for $\rho(e)$ (HDC binding under non-commutative random shifts; see Appendix C).

Cochain complex. Set

$$C^0 = \bigoplus_{v \in V} \mathcal{F}(v), \qquad C^1 = \bigoplus_{e \in E} \mathcal{F}(e), \qquad C^2 = \bigoplus_{\Delta} \mathcal{F}(\Delta).$$

The coboundary $\delta_0 : C^0 \to C^1$ is defined edge-wise by $(\delta_0 x)_e = \mathcal{F}_{v \blacktriangleleft e} x_v - \mathcal{F}_{u \blacktriangleleft e} x_u$. The coboundary $\delta_1 : C^1 \to C^2$ is the alternating sum of edge values around each triangle $\Delta = (e_1, e_2, e_3)$.

Cohomology.

$$H^0(\mathcal{F}) = \ker \delta_0, \qquad H^1(\mathcal{F}) = \ker \delta_1 / \operatorname{im} \delta_0.$$

Standard arguments (see e.g. Hansen & Gebhart, [6]) verify $\delta_1 \delta_0 = 0$, so H^1 is well-defined.

Implementation and complexity. We compute $\dim H^0$ as the multiplicity of the zero eigenvalue of the sheaf Laplacian $L = \delta_0^\top \delta_0 + \delta_1 \delta_1^\top$ within tolerance 10^{-6} via dense symmetric eigensolve (`numpy.linalg.eigvalsh`), and $\dim H^1 = \dim \ker \delta_1 - \operatorname{rank} \delta_0$ via dense rank computation on the boundary matrices. With n vertices and stalk dimension d, this is $O((nd)^3)$ time and $O((nd)^2)$ space; tractable for the type-graph view we use in our experiments ($n \leq 100$ Olog objects per ontology), but not for the full entity-level KG. We note the implementation uses sparse Krylov diffusion (`scipy.sparse.linalg.expm_multiply`) for query-time heat propagation, but cohomology computation itself remains dense. A sparse sheaf-cohomology algorithm via simultaneous Krylov-subspace methods on δ_0 and δ_1 is a natural extension we leave to future work; the cleaner alternative (computing the alternating sum $\chi = \dim H^0 - \dim H^1 + \dim H^2$ via Euler characteristic in $O(n+m)$) yields only the alternating sum, not individual Betti numbers, when restriction maps are non-trivial.

B Proof of Theorem 1

We restate Theorem 1 formally.

Theorem 2 (Conflict Sensitivity, full statement)**.** *Let $\mathcal{F}$ be the cellular sheaf on $G = (V, E)$ defined in Appendix A. A* conflict injection *of size k is a set $E^* = \{e_1^*, \ldots, e_k^*\}$ of additional edges with restriction maps $\{(\mathcal{F}^*_{u_i \blacktriangleleft e_i^*}, \mathcal{F}^*_{v_i \blacktriangleleft e_i^*})\}$ chosen so that for each i, no global section $x \in \ker \delta_0$ of $\mathcal{F}$ satisfies $\mathcal{F}^*_{v_i \blacktriangleleft e_i^*} x_{v_i} = \mathcal{F}^*_{u_i \blacktriangleleft e_i^*} x_{u_i}$. Let $\mathcal{F}'$ denote the resulting sheaf on $G' = (V, E \cup E^*)$. Then*

$$\dim H^1(\mathcal{F}') \;\geq\; \dim H^1(\mathcal{F}) + c, \tag{3}$$

where c is the dimension of the subspace of $\bigoplus_{i=1}^k \mathcal{F}(e_i^*)$ spanned by the conflict residuals $r_i = \mathcal{F}^*_{v_i \blacktriangleleft e_i^*} x_{v_i} - \mathcal{F}^*_{u_i \blacktriangleleft e_i^*} x_{u_i}$ evaluated on any δ_0-section, modulo $\operatorname{im} \delta_0' + \operatorname{im} \delta_2'^\top$. Trivially, $c \leq k$.

Proof. The cochain spaces decompose as

$$C^{0\prime} = C^0, \qquad C^{1\prime} = C^1 \oplus \bigoplus_{i=1}^k \mathcal{F}(e_i^*), \qquad C^{2\prime} = C^2 \oplus C^2_*$$

where C^2_* collects any new 2-cells closed by injected edges (possibly empty).

Write $\delta_0' : C^{0\prime} \to C^{1\prime}$ in block form:

$$\delta_0' \;=\; \begin{pmatrix} \delta_0 \\ A \end{pmatrix},$$

where $A : C^0 \to \bigoplus_i \mathcal{F}(e_i^*)$ is the restriction-map contribution from the conflict edges, $Ax = (r_1(x), \ldots, r_k(x))$.

The conflict hypothesis is precisely $A|_{\ker \delta_0} \neq 0$, i.e. there is no $x \in \ker \delta_0$ with $Ax = 0$ unless we shrink to a strict subspace. Equivalently, $\ker \delta_0' \subsetneq \ker \delta_0$, and $\dim \ker \delta_0' = \dim \ker \delta_0 - r_A$ where $r_A = \operatorname{rank} A|_{\ker \delta_0}$.

By rank–nullity for δ_0':

$$\operatorname{rank} \delta_0' \;=\; \dim C^{0\prime} - \dim \ker \delta_0' \;=\; \operatorname{rank} \delta_0 + r_A.$$

For H^1:

$$\dim H^1(\mathcal{F}') \;=\; \dim \ker \delta_1' - \operatorname{rank} \delta_0'.$$

We have $\dim \ker \delta_1' \geq \dim \ker \delta_1 + k - \dim \operatorname{im} \delta_2'^{\top}$ because each new edge contributes a new generator to $C^{1\prime}$, which is killed by δ_1' unless it participates in a new 2-cell relation in C_*^2.

Combining,

$$\begin{aligned} \dim H^1(\mathcal{F}') \;&\geq\; \dim \ker \delta_1 + k - \dim \operatorname{im} \delta_2'^{\top} - \operatorname{rank} \delta_0 - r_A \\ &=\; \dim H^1(\mathcal{F}) + (k - r_A - \dim \operatorname{im} \delta_2'^{\top}). \end{aligned}$$

Setting $c = k - r_A - \dim \operatorname{im} \delta_2'^{\top}$ gives inequality (3). The bound $c \leq k$ is immediate. □

Empirical alignment. Our experiment (Table 1) gives $\Delta H^1 = +53$ for $k = 76$, hence $c/k \approx 0.70$. Roughly 70% of injected conflicts contribute linearly independent classes to H^1; the remainder are absorbed by $\operatorname{im} \delta_0$ (i.e. correctable by a global section adjustment) or by 2-cell boundaries.

C Implementation Details

The system has four components, each a separate module in the public code release (Appendix F).

HDC encoder. Entities and relations are encoded as 4096-dimensional bipolar (± 1) hypervectors [7]. Relation application uses non-commutative binding: each relation r is assigned a fixed random circular shift σ_r, so that $\sigma_r(h) \neq \sigma_r^{-1}(h)$ and a directed edge $h \xrightarrow{r} t$ is distinguishable from its reverse. The circular-shift (permutation) binding follows the holographic reduced representation family [9].

Typed attention (B). The (B) layer adds a precomputed mask M to the pre-softmax attention scores, with $M_{ij} = 0$ if $\tau(j) \in \text{Reach}_{\mathcal{O}}(\tau(i))$ and $M_{ij} = -\infty$ otherwise (in practice -10^9). The mask is built once per Olog from the transitive closure of the morphism set.

Constrained decoder (D). At each generation step the decoder queries the proof object for the recommended step (t, r, t'), constructs the admissible-label set $\mathcal{A}(t, r, t')$, and applies a logit mask of $-\infty$ to every vocabulary entry outside it. The decoder is stateless across steps apart from the proof object and composes with (B) without interference.

Table 6: HDC/Sheaf hyperparameter sweep on FB15K-237, 30 epochs per configuration, ranked by MRR. Diffusion time is a query-time heat-kernel parameter and does not enter training; rows 3–6 (identical except for diffusion time $\in \{0.5, 1.5, 3.0, 5.0\}$) produce identical ranking metrics, confirming that retrieval accuracy is decoupled from the diffusion parameter. The 30-epoch cap depresses absolute MRR relative to the 50-epoch headline run of Table 5 (0.346); the sweep is for relative comparison only.

Rank	HDC Dim / LR / Diff. Time	MRR	Hits@10	Note
1	1024 / 0.001 / 1.5	**0.2284**	0.3966	best MRR
2	2048 / 0.001 / 1.5	0.2276	**0.4024**	best Hits@10
3–6	4096 / 0.001 / {0.5–5.0}	0.2264	0.3989	diffusion-invariant
7	4096 / 0.003 / 1.5	0.2259	0.3958	
8	4096 / 0.0001 / 1.5	0.2257	0.3948	lr too low
9	4096 / 0.010 / 1.5	0.2097	0.3742	lr instability
10	8192 / 0.001 / 1.5	0.2096	0.3807	underfit at 30 ep.

Sheaf cohomology. The cohomology module builds the boundary operators δ_0, δ_1 from the restriction maps of Appendix A (identity on the head side, relation embedding on the tail side) and computes $\dim H^1$ by dense rank on the boundary matrices for the type-graph view, and by the $O(n+m)$ sparse connected-components route for the entity-graph consistency score of Table 4.

Training configuration. The attention ablation (Table 2, Figure 2) uses a two-head, 64-dimensional attention model trained 300 epochs with Adam (learning rate 0.003) on next-type prediction over 23 types drawn from four hand-built domain Ologs (business, academic, healthcare, e-commerce). The HDC/Sheaf link-prediction model (Table 5) uses 4096-dimensional embeddings trained 50 epochs at batch size 512. All runs use seed 42.

D Hyperparameter Sweep

We ran a 10-configuration sweep over HDC embedding dimension, learning rate, and query-time diffusion time on FB15K-237, capped at 30 epochs per configuration for rapid iteration (Table 6).

Two findings. First, at a 30-epoch budget the lower dimensions (1024–2048) converge faster and edge out 4096+, while 8192 underfits — larger hypervector spaces need more epochs to populate. Second, query diffusion time has no effect on ranking metrics: it governs only the heat-kernel spread used for path retrieval, not the embedding geometry. Learning rate 0.001 is the stable choice; 0.01 induces optimization instability and 0.0001 is too conservative.

E Cyclic Olog Stress Test

The cyclic stress test (Table 3, second row) takes the 7-type e-commerce Olog and adds back-edges (Delivery $\to$ Customer, Item $\to$ Customer) that close the otherwise acyclic schema into a cycle. The effect is structural: with the cycle present, every type can reach every other type, so the reachability relation $\text{Reach}_{\mathcal{O}}$ becomes the universal relation (reachability rises from 34.7% of type pairs to 100%).

Because (B′) admits any next label whose target type is reachable, a universal reachability relation makes the (B′) mask admit everything: (B′) GOOD soundness falls from 65.0% to 8.0% and (B′) BAD from 3.9% to 0.0%, both within roughly one point of the unconstrained (A) baseline

(6.9% / 0.0%). The reachability mask is, at that point, doing no work. The (D) constrained decoder is unaffected — it masks to the direct-edge admissible set $\mathcal{A}(t, r, t')$, which does not grow when the Olog gains a cycle — and remains at 100% trajectory soundness on both priors. This is the predicted topological limit: reachability masking degrades to no-constraint as Olog connectivity grows, whereas direct-edge masking is invariant to topology.

F Compute and Reproducibility

Hardware and runtime. The HDC/Sheaf link-prediction model trains on a single A100-40GB GPU (Modal), 50 epochs at batch size 512, ≈30 s/epoch (≈25 min end-to-end). The attention ablation runs on CPU, ≈5 min per model for 300 epochs. The loci and cyclic-stress trajectory experiments run on CPU in seconds ($N = 1000$ trajectories per cell). Type-graph sheaf cohomology is a dense $O((nd)^3)$ computation but tractable at $n \leq 100$; the entity-graph consistency score uses the $O(n + m)$ sparse route.

Code and seeds. The implementation — HDC encoder, typed attention, constrained decoder, and the `compute_cohomology` routine — is available at `https://github.com/oasis-main/alignment_research` under `ontological_embeddings/`. All experiments use seed 42; the attention ablation is reported single-seed and a multi-seed re-run is noted as pending in §5. Panel (a) of Figure 2 is computed deterministically from the Olog reachability closure; panel (b) is read directly from the attention-ablation results of Table 2.

G (B) vs (D) — Empirical Detail and Open Question

Empirical setup. (B′), the sampling-time projection of (B), admits a next label ℓ if its target type is reachable from the current state under the Olog's morphism closure. We compare (A) unconstrained, (B′) reachability-projected, (C) in-FFN compilation (applicable only on functional sub-fragments), and (D) constrained decoding on synthetic trajectories from the e-commerce Olog under both GOOD and BAD label priors. Trajectory soundness is the fraction of generated trajectories that decode to admissible $\delta_{\mathcal{O}}$-paths.

Result. See Table 3 in the main body for the full numbers. Headlines: under acyclic Olog, (B′) GOOD reaches 65.0% trajectory soundness vs (A) GOOD's 48.1% and (D)'s 100%; under BAD prior (B′) sits at 3.9% vs (A)'s 4.2%, no meaningful improvement. Under cyclic Olog (Delivery → Customer added), reachability becomes 100% universal and (B′) collapses to 8.0% GOOD / 0.0% BAD, within a point of (A); (D) remains 100%.

Open question (the curiosity case). If (D) alone achieves 100% trajectory soundness, what does (B) buy when (D) is also present? Three hypotheses, none yet tested: (i) *representation prior:* (B) restricts information flow during training, so the residual stream carries cleaner type information; (D) alone leaves the residual unconstrained even if outputs are valid. (ii) *efficiency:* (B) reduces effective attention bandwidth during forward pass, potentially saving compute on long sequences in exchange for the per-layer mask cost. (iii) *robustness under decoder failure:* if (D) is removed (e.g. during research probing), a (B)-trained model degrades more gracefully than a (B)-free one. A four-cell ablation (A / B / D / B+D) trained from scratch on the same data, with both downstream task accuracy and trajectory soundness measured at convergence, would discriminate. We leave this

to follow-up work; the present paper reports (B) and (D) separately and does not claim a Pareto win for (B)+(D) without it.

Implementation pointers. (B) is the typed attention layer described in §4.2; the production reference is `OntologicalAttention` in `ontological_attention.py` (line 661, `run_slot_ablation`). (D) is implemented in `ConstrainedDecoder` of `proof_guided_generation.py` (line 475). The decoder constructs the admissible-label set per generation step from the proof object and applies a logit mask of $-\infty$ to non-admissible vocabulary entries. Sheaf cohomology is computed by the `compute_cohomology` method of `ontology_sheaf.py` (line 219), which is now vendored into the public repository.